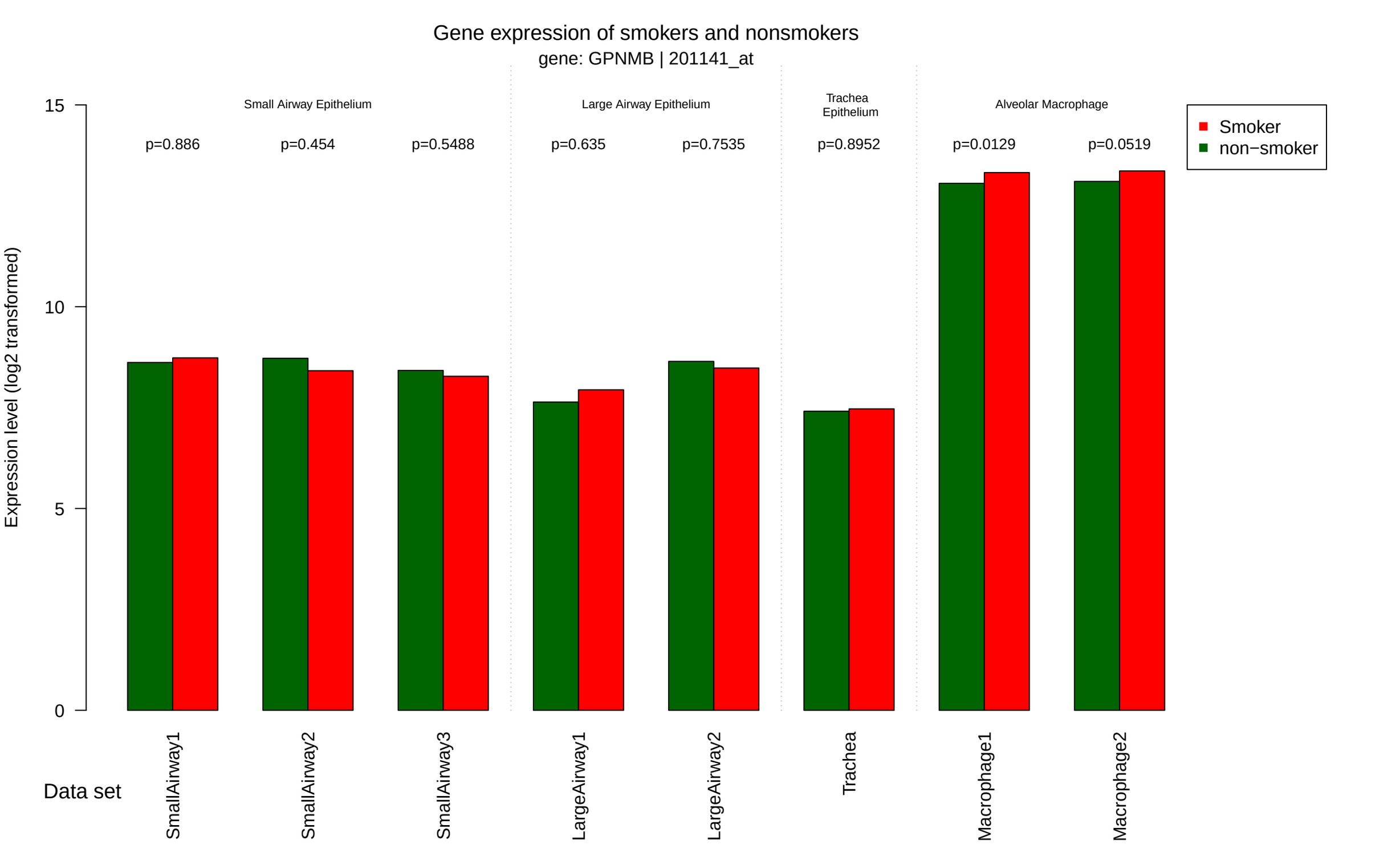
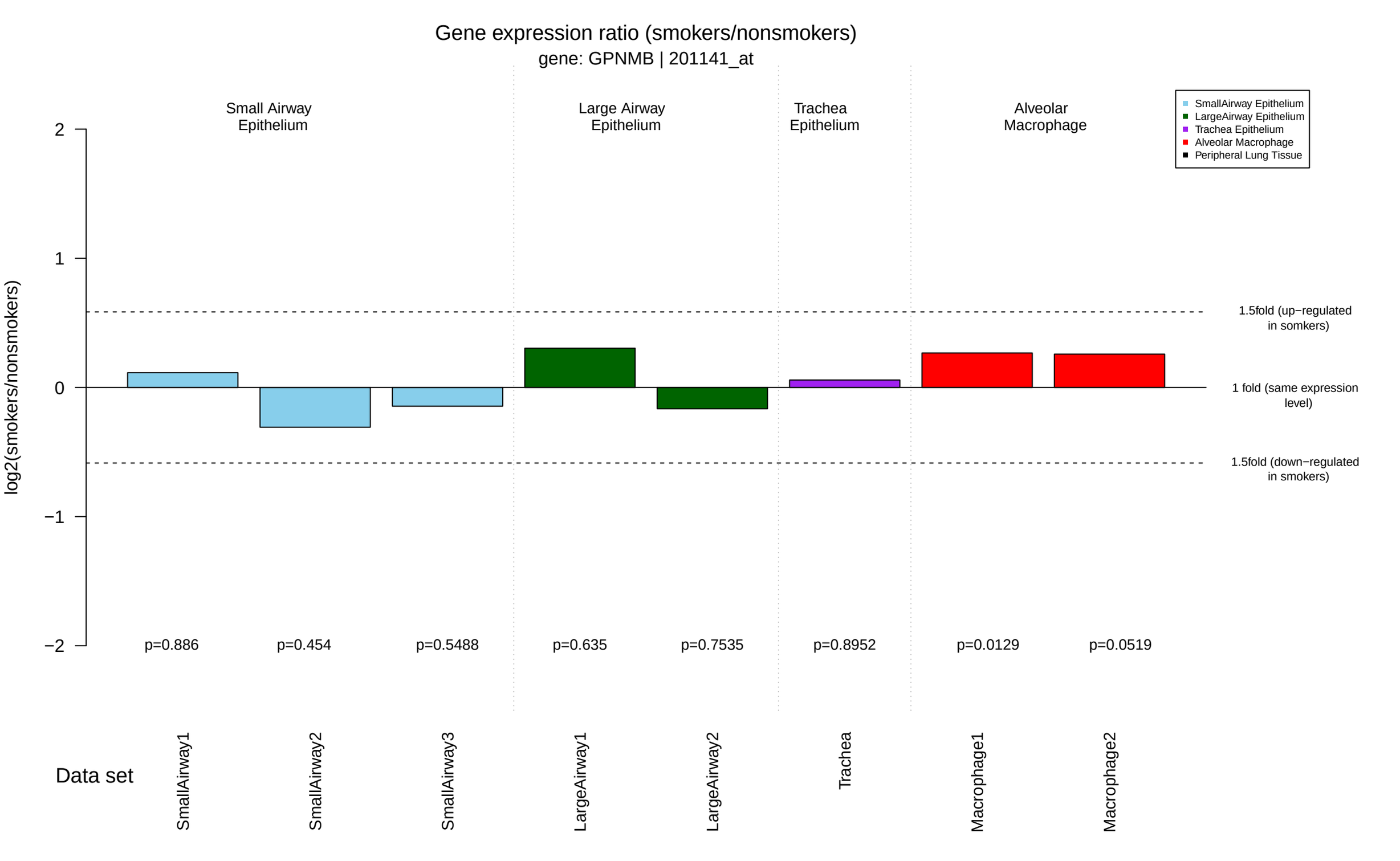
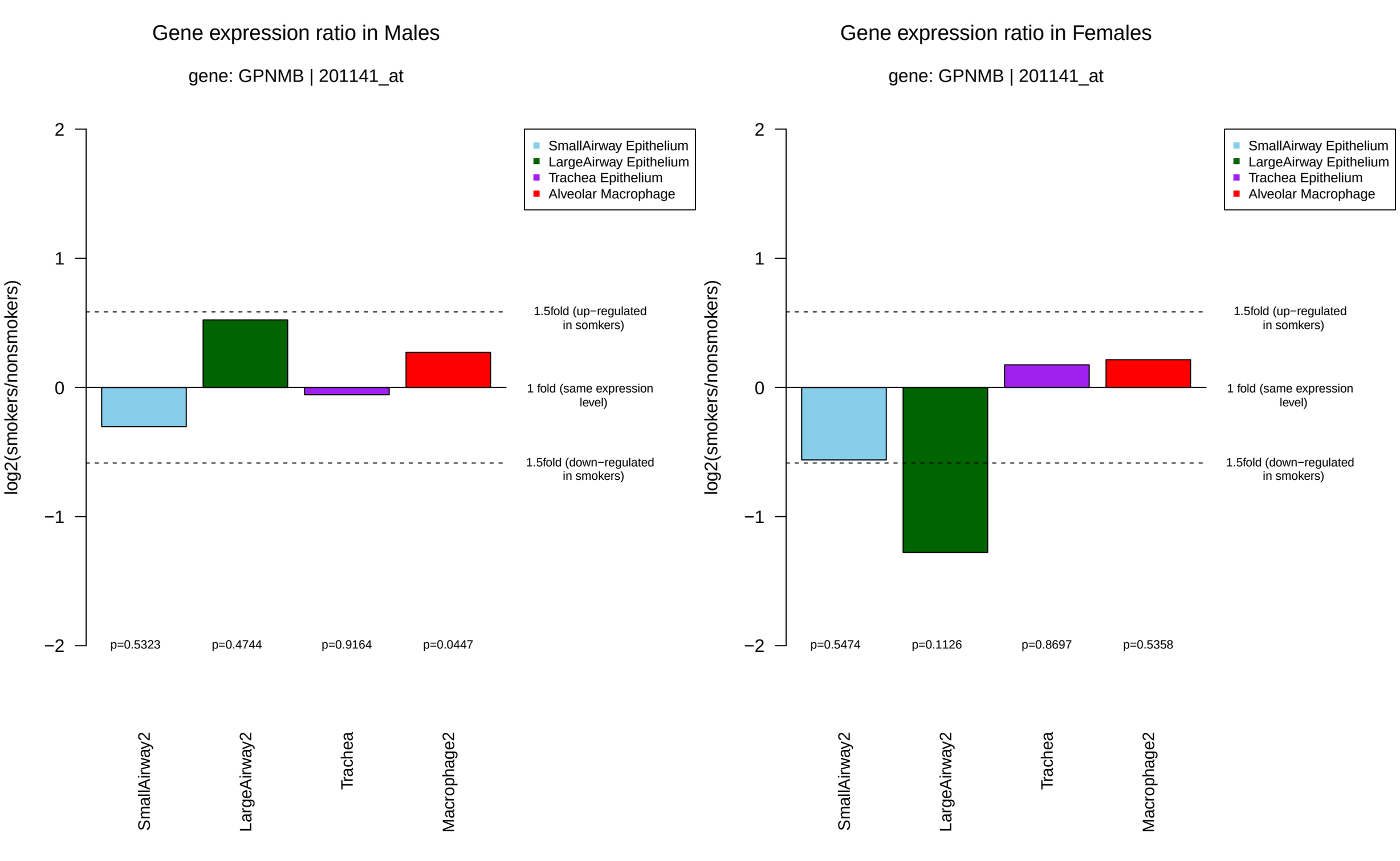
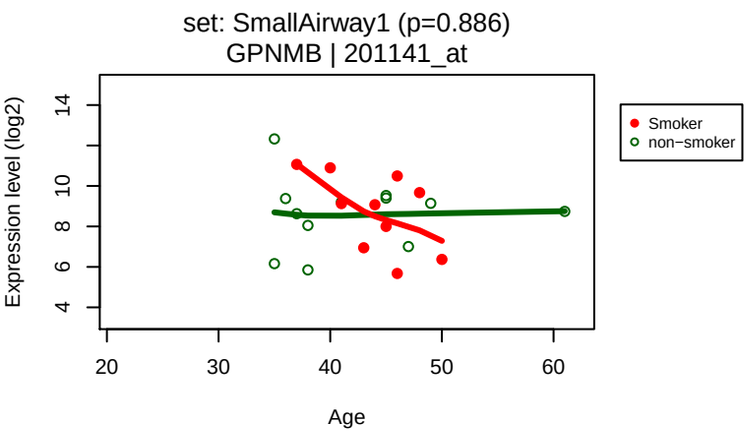
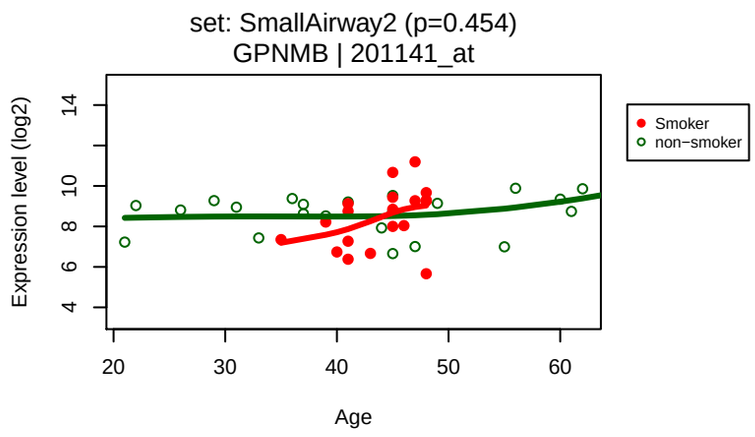
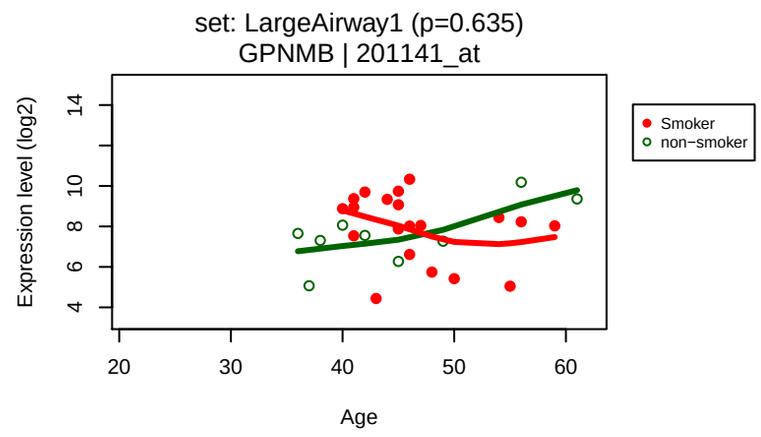

Gene expression of smokers and nonsmokers
gene: GPNMB | 201141_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.886
p=0.454
p=0.5488
p=0.635
p=0.7535
p=0.8952
p=0.0129
p=0.0519
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

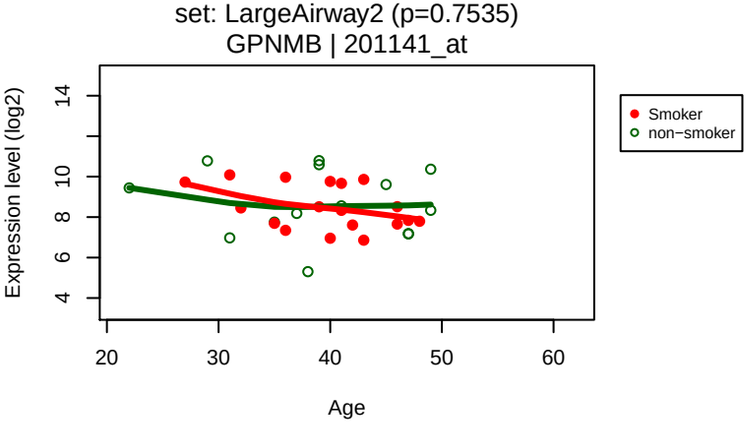

Gene expression ratio (smokers/nonsmokers)
gene: GPNMB | 201141_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.886
p=0.454
p=0.5488
p=0.635
p=0.7535
p=0.8952
p=0.0129
p=0.0519
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

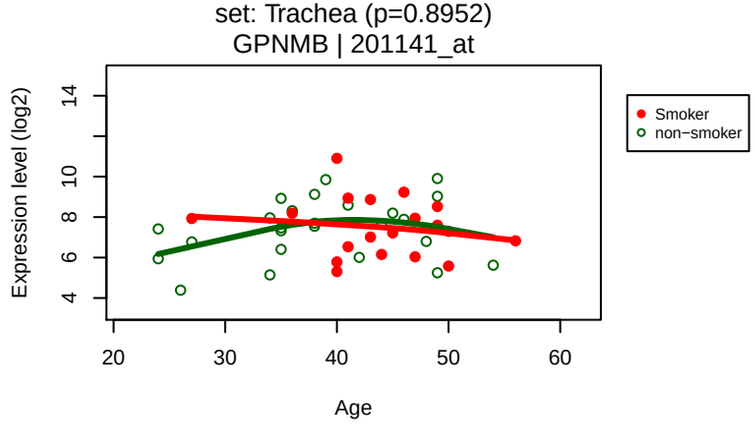

Gene expression ratio in Males
gene: GPNMB | 201141_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5323
p=0.4744
p=0.9164
p=0.0447
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPNMB | 201141_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5474
p=0.1126
p=0.8697
p=0.5358
SmallAirway2
LargeAirway2
Trachea
Macrophage2

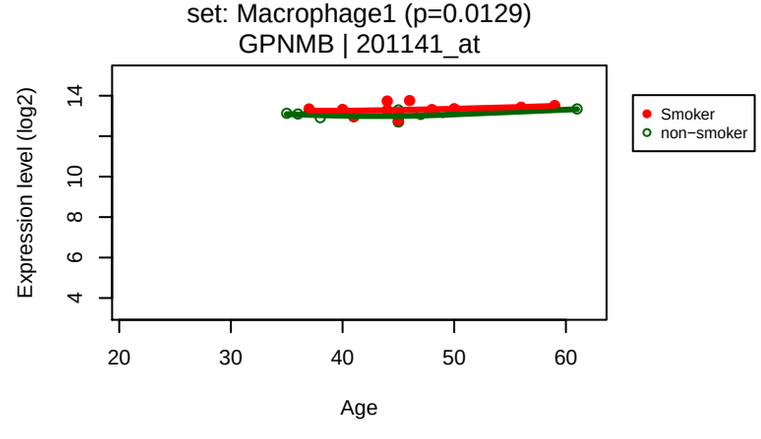

set: SmallAirway1 (p=0.886)
GPNMB | 201141_at
Expression level (log2)
Age
Smoker
non−smoker

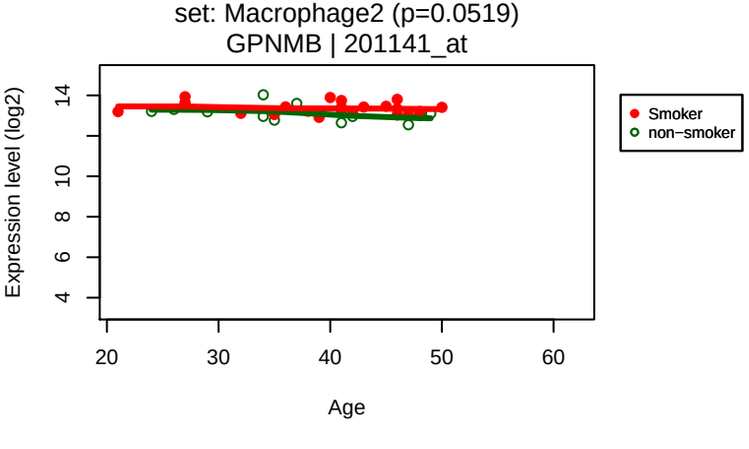

set: SmallAirway2 (p=0.454)
GPNMB | 201141_at
Expression level (log2)
Age
Smoker
non−smoker

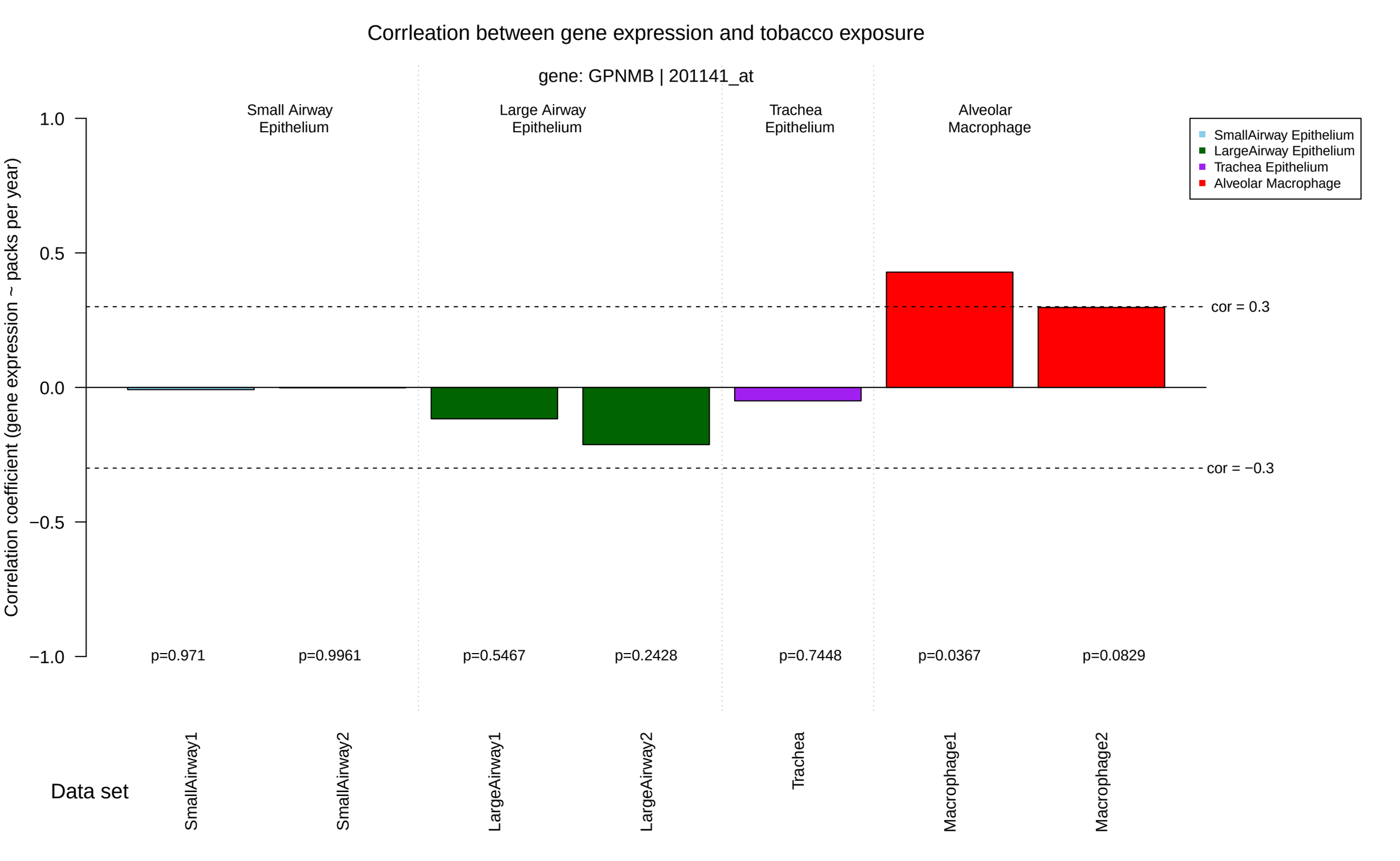

set: LargeAirway1 (p=0.635)
GPNMB | 201141_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7535)
GPNMB | 201141_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8952)
GPNMB | 201141_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0129)
GPNMB | 201141_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0519)
GPNMB | 201141_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPNMB | 201141_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.971
p=0.9961
p=0.5467
p=0.2428
p=0.7448
p=0.0367
p=0.0829
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2